

■■ SafeRoute — Technical Architecture & System Guide

AI-Powered Women's Safety Navigation Engine | Live Production Architecture

1. Executive Summary & Problem Statement

The Problem: Standard navigation platforms (Google Maps, Waze, Apple Maps) optimize exclusively for shortest distance or fastest travel time. They frequently guide pedestrians and drivers through dark, unlit back-alleys, isolated industrial zones, or high-crime corridors because it saves 60 seconds.

The SafeRoute Solution: SafeRoute evaluates street lighting sensors, police station proximity, nightfall risk multipliers, commercial footfall ('eyes on the street'), and crowd-sourced hazard reports to strictly prioritize **Women's Safety First**.

2. Geocoding & OSRM Road Network Engine

- **Geocoding (Text to GPS):** SafeRoute converts user location inputs into exact spatial coordinates via Nominatim OpenStreetMap API (e.g., Hitech City: 17.4435° N, 78.3772° E; Banjara Hills: 17.4150° N, 78.4350° E).
- **OSRM Road Geometry Query:** Queries OSRM API with GeoJSON enabled (<https://router.project-osrm.org/route/v1/driving/...>) to return array of coordinate waypoints along actual physical streets.

3. The Haversine Spatial Distance Formula

SafeRoute executes the spherical **Haversine Distance Formula** to measure exact spatial distances on Earth's curved surface between moving users, road waypoints, police stations, and hazard pins:

$$d = 2R \cdot \arcsin(\sqrt{ \sin^2(\Delta\phi/2) + \cos(\phi_1) \cdot \cos(\phi_2) \cdot \sin^2(\Delta\lambda/2) })$$

- **Police Proximity:** Scans for police stations & She Team kiosks within **1.0 km**.
- **Hospital Proximity:** Identifies 24/7 emergency medical centers within **1.5 km**.
- **Hazard Exposure Radius:** Checks for reported snatching/harassment pins within **500 meters**.
- **Off-Route 200m Safety Fence:** Triggers instant deviation warning if user moves >200m away from planned safe route.

4. The 5-Factor Mathematical Safety Scoring Algorithm (0–100)

SafeRoute starts every candidate route with **100 Base Points** and applies the mathematical scoring formula:

$$\text{Final Score} = 100 - \text{Unlit Penalty} + \text{Police Bonus} - (\text{Hazard Penalty} \times \text{Night Multiplier}) + \text{Footfall Bonus}$$

Safety Factor	Weight	Operational Logic
■ Street Lighting	30%	OSM lit=yes tags + Municipal lighting hierarchy
■ Police & Emergency	25%	Haversine distance <= 1.0 km adds +15 bonus pts
■■ Hazard Incident Pins	20%	Snatching/harassment pins <= 500m deduct -20 to -35 pts
■ Nightfall Risk Shift	15%	8 PM to 5 AM applies 1.4x hazard penalty multiplier
■ Commercial Footfall	10%	Active shop/metro density per 100m adds +10 pts

5. Street Lighting & Commercial Footfall Intelligence

- **3 Lighting Layers:** Combines OpenStreetMap infrastructure tags (lit=yes), municipal road hierarchy (commercial highways = 85-95% lit), and crowd-sourced 'Unlit Streetlight' hazard reports.
- **Commercial Footfall ('Eyes on the Street'):** Based on Jane Jacobs' urban safety principle. Streets with active retail stores, cafes, ATMs, and metro stations provide natural surveillance and emergency shelter access (+10 bonus pts).

6. Zero-Delay Emergency SOS Pipeline (n8n + Twilio)

- **CORS-Free Architecture:** Native HTML anchor triggers target a hidden background iframe, guaranteeing 0ms execution without browser CORS preflight blocking.
- **Cloud Phone Call Automation:** Triggers n8n production webhook (<https://pranav3010.app.n8n.cloud/webhook/sos-trigger>), which executes Twilio REST API to place a live voice call to +916300863028.
- **Multilingual Voice SOS Engine:** Web Speech API continuously listens for panic triggers ('HELP ME', 'SAVE ME', 'BACHAO', 'MADAD KARO').

7. Master Hackathon Presentation Pitch Script

'SafeRoute is a comprehensive women's safety navigation platform. Unlike standard navigation apps that only optimize for travel speed, SafeRoute combines real-time OSRM road geometry with the spatial Haversine distance formula to evaluate street lighting, police station proximity, nightfall risk multipliers, commercial footfall, and crowd-sourced hazard reports—strictly prioritizing safety over shortest distance. In an emergency, SafeRoute features zero-delay one-tap and hands-free voice SOS that triggers an n8n cloud automation pipeline—placing live phone calls via Twilio REST API to emergency contacts while broadcasting real-time GPS location tracking.'